

Processamento Digital de Sinais

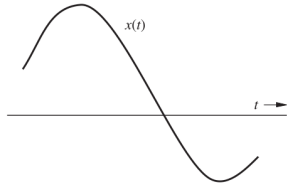
Prof. Dr. Paulo Rogério de Almeida Ribeiro

Coordenação do Curso de Engenharia da Computação

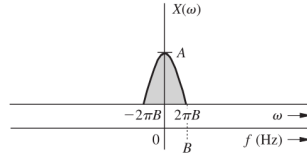
Amostragem (recuperação do sinal)



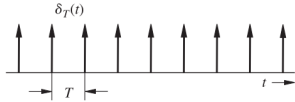
Processo de amostragem: tempo e frequência (Lathi, 2006)



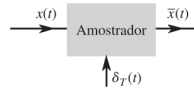
(a)



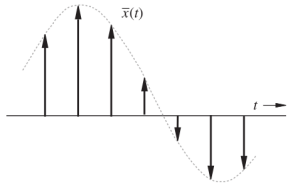
(b)



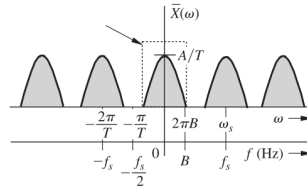
(c)



(d)

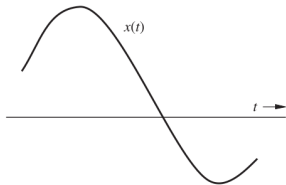


(e)

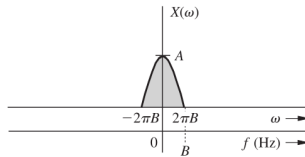


(f)

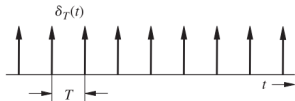
Processo de amostragem com recuperação do sinal (Lathi, 2006)



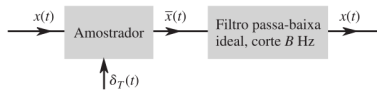
(a)



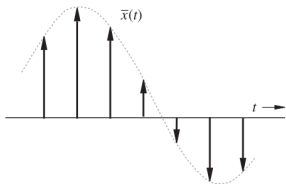
(b)



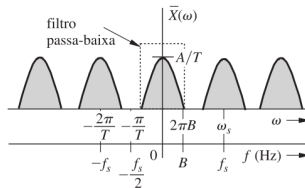
(c)



(d)

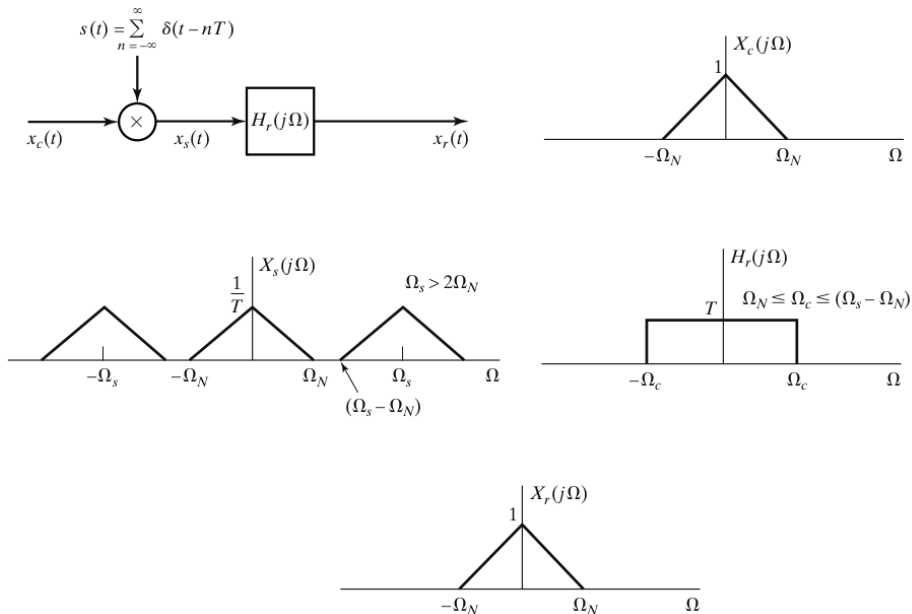


(e)

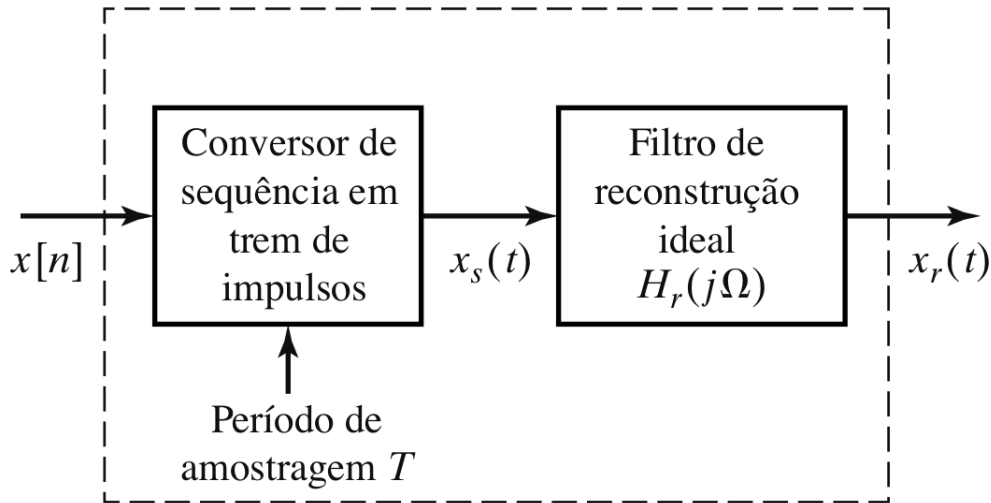


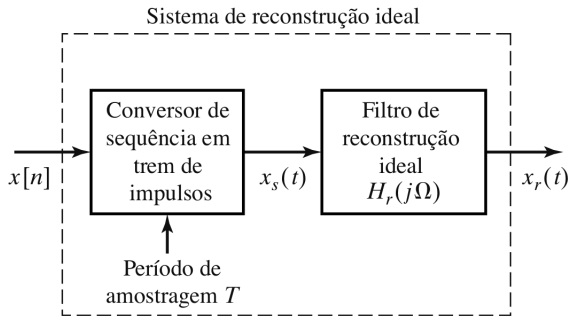
(f)

Processo de amostragem com recuperação do sinal (Oppenheim and Schafer, 2013)

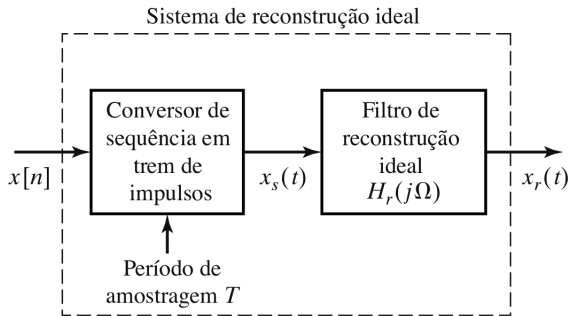


Sistema de reconstrução ideal





$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT)$$

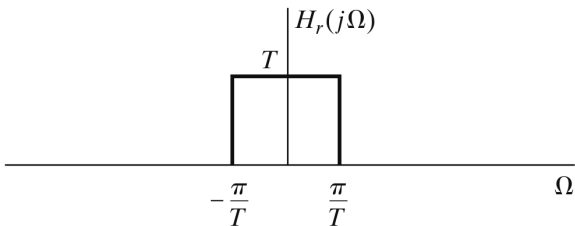


$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT)$$

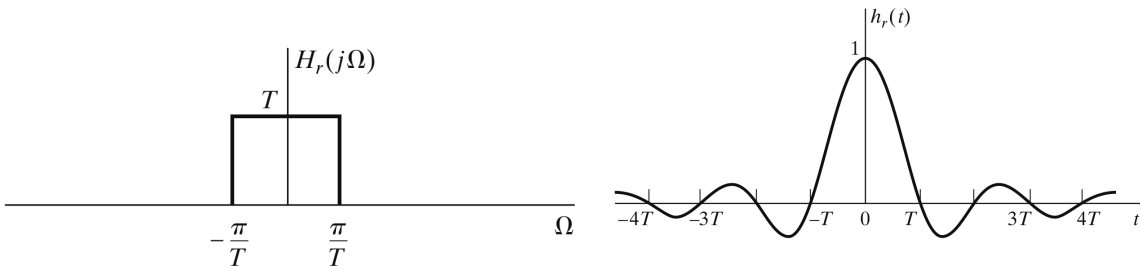
Passando pelo filtro (H_r):

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t - nT)$$

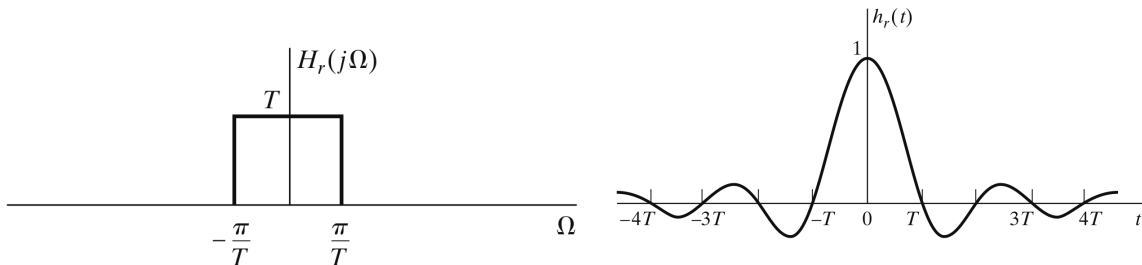
Filtro passa-baixa para reconstrução do sinal (Oppenheim and Schaffer, 2013)



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Frequência de corte: $\omega_c = \frac{\omega_s}{2} = \frac{\pi}{T}$

Voltemos no código efeito de T e adicionamos um filtro passa baixa.

Filtro no domínio do tempo

$$h_r(t) = \frac{\text{sen}\left(\frac{\pi t}{T}\right)}{\frac{\pi t}{T}}$$

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$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\text{sen}\left[\frac{\pi(t-nT)}{T}\right]}{\frac{\pi(t-nT)}{T}}$$

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Propriedades da sinc - $h_r(t)$:

- $h_r(0) = 1$

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- $h_r(0) = 1$
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Sabendo, da amostragem, que $x[n] = x_c(nT)$ e usando as propriedades temos:

$$x_r(mT) = x_c(mT)$$

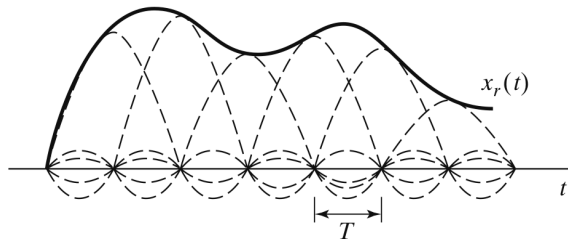
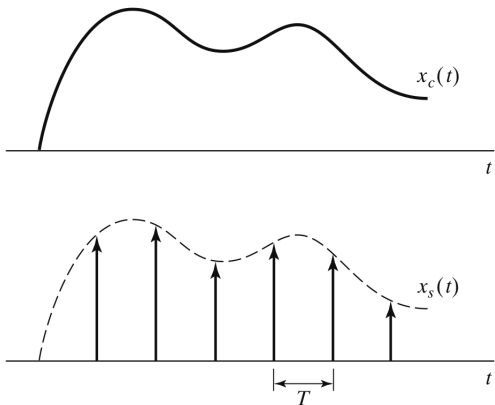
Reconstrução do sinal

$$x_r(mT) = x_c(mT)$$

O sinal reconstruído (x_r) tem os mesmos valores que o original (contínuo, x_c) nos instantes de amostragem T

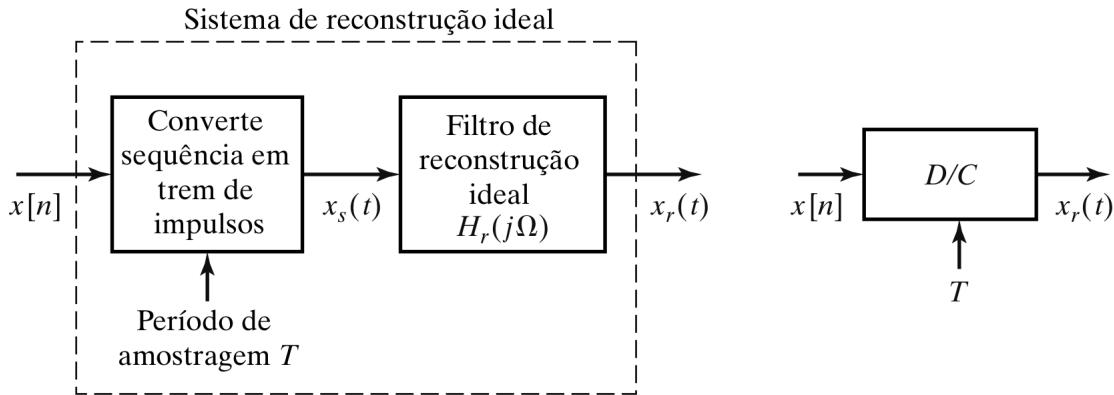
Reconstrução do sinal (Oppenheim and Schafer, 2013)

Adicionalmente, o filtro passa-baixa interpola (com a convolução) entre os impulsos para calcular $x_r(t)$



Reconstrução ideal do sinal de banda limitada

Representação equivalente: conversor D/C ideal (Oppenheim and Schaffer, 2013)



Aspectos práticos da amostragem e recuperação

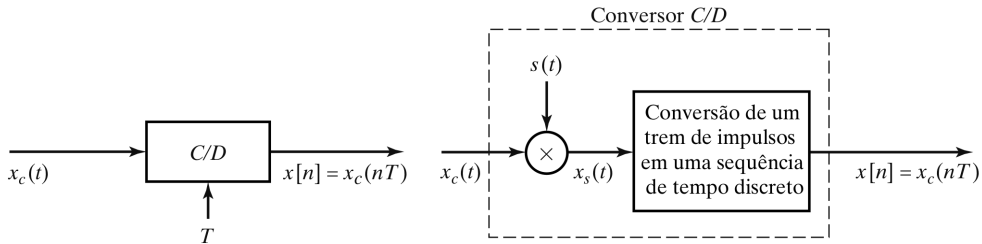
Aspectos práticos da amostragem e recuperação

- Amostragem: Se o sinal não for de banda limitada?

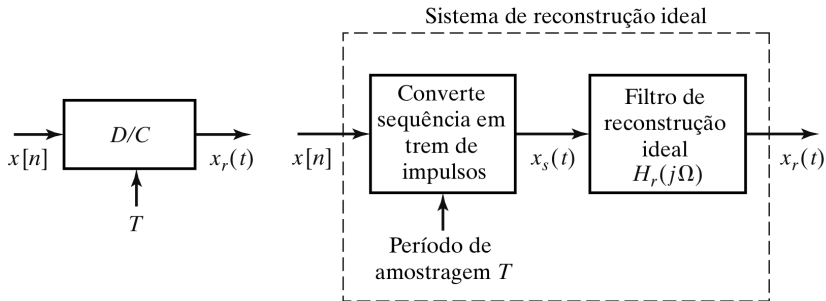
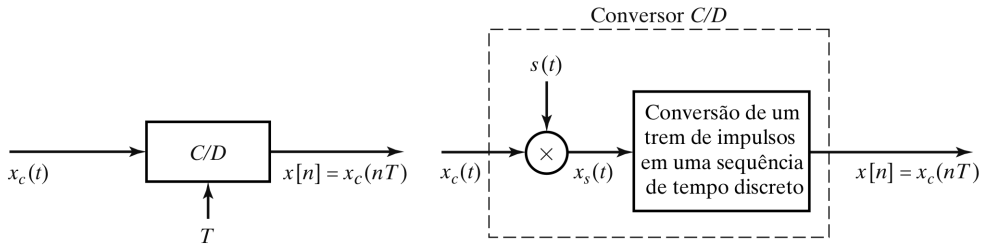
Aspectos práticos da amostragem e recuperação

- Amostragem: Se o sinal não for de banda limitada?
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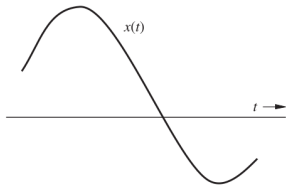
Recapitulando amostragem e recuperação (Oppenheim and Schafer, 2013)



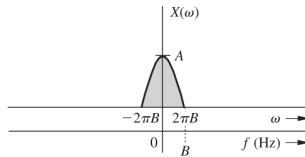
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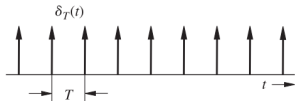
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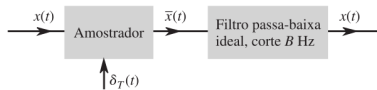
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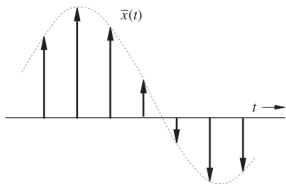
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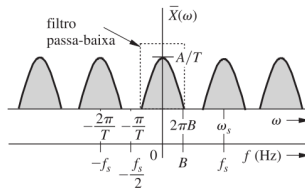
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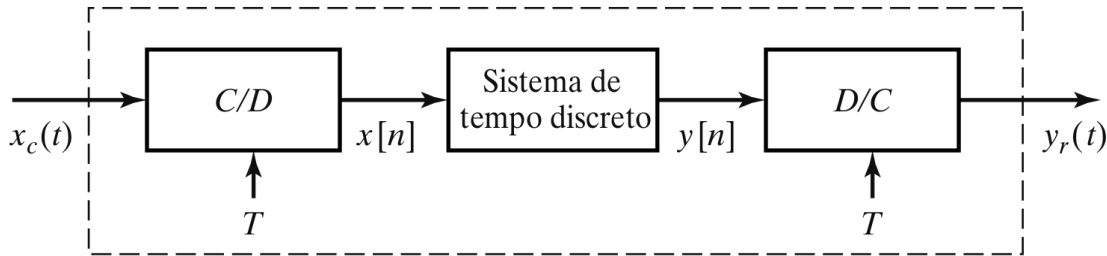


(e)



(f)

Com essas operações podemos realizar: Processamento em tempo discreto de sinais de tempo contínuo



(Oppenheim and Schaffer, 2013)

Referência bibliográfica

Lathi, B. (2006). *Sinais e Sistemas Lineares*. Bookman, 2 edition.

Oppenheim, A. V. and Schafer, R. W. (2013). *Processamento em tempo discreto de sinais*. Pearson, 3 edition.